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10

$$a) \frac{dm_{sys}}{dt} = m_i - m_e$$

$$m_i = 10 \text{ kg/min}, m_e = 15 \text{ kg/min}$$

$$\Rightarrow \frac{dm_{sys}}{dt} = 10 \frac{\text{kg}}{\text{min}} - 15 \frac{\text{kg}}{\text{min}} = -5 \frac{\text{kg}}{\text{min}}$$

$$\Rightarrow \int dm_{sys} = -5 \int dt \Rightarrow \boxed{m_{sys}(t) = -5t}$$

$$m_{sys} \Big|_{100}^{m_{sys}} = -5 \Big|_0^t \quad \text{at } t=0, m_{sys}(0) = 100 \text{ kg}$$

$$\Rightarrow m_{sys} - 100 = -5t \Rightarrow \boxed{m_{sys} = -5t + 100}$$

$$b) \text{ after at } t = 10 \text{ min} \Rightarrow m_{sys}(10) = -5(10) + 100$$

$$\Rightarrow \boxed{m_{sys} = 50 \text{ kg}} \text{ after 10 minutes}$$

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$$\begin{aligned}
 c) \quad \frac{dw_{\text{sys}}}{dt} &= 0.1(10) - 15w \\
 \Rightarrow \frac{d\{w(-5t+100)\}}{dt} &= 1-15w \\
 \Rightarrow \frac{d\{-5t+100\}}{dt} \{w\} &= 1-15w \\
 \Rightarrow -5w + (-5t) \frac{dw}{dt} + 100 \frac{dw}{dt} &= 1-15w \\
 \Rightarrow \frac{dw}{dt} \{-5t+100\} &= 1-10w \Rightarrow \frac{dw}{1-10w} = \frac{dt}{-5t+100} \\
 \Rightarrow \frac{-1}{10} \ln|1-10w| &= \frac{-1}{5} \ln|-5t+100| \Rightarrow \ln|1-10w| \Big|_0^w = 2 \ln|-5t+100| \Big|_{100}^t \\
 \Rightarrow \ln|1-10w|_0 - \ln 5 &= 2 \ln|-5t+100| - 2 \ln 100 \\
 \Rightarrow \ln \left| \frac{1-10w}{5} \right| &= 2 \ln \left| \frac{-5t+100}{100} \right| \\
 \Rightarrow \left| \frac{1-10w}{5} \right| &= e^{\frac{2 \ln|-5t+100|}{100}} = \left(\frac{-5t+100}{100} \right)^2 \\
 \Rightarrow \left| 1-10w(t) \right| &= 5 \left(\frac{-5t+100}{100} \right)^2 \\
 \text{at } t=10 \text{ minutes} \Rightarrow \left| 1-10w(10) \right| &= 5 \left(\frac{-50+100}{100} \right)^2 \\
 \Rightarrow w(t=10) &= 0.225 \quad \checkmark \text{ Perfect!}
 \end{aligned}$$

The weight fraction of A in the tank after 10 minutes is $w(t=10) = 0.225$

Great Job!!!